As before, it is easy to construct a two-step estimator based on the sample analogue of the result in the last proposition. After some algebra, the result in equation (10) can be obtained applying Proposition 3.1 to the case when $g(X_k, \theta)$ is constant.

3.2. Some extensions and particular cases

This section contains extensions to repeated cross sections, and multilevel treatments. It also discusses the case of selection on observables.

3.2.1. Repeated cross sections. Often, a random sample with repeated outcomes is not available. In such a case, repeated cross-section data-sets (pre-treatment and post-treatment) may be used to construct DID estimators. However, the use of repeated cross sections for DID presents some issues of data availability. First, treatment status (in the post-treatment period) must be known for the individuals in the pre-treatment sample. This requirement is satisfied, for example, if treatment exposure can be determined from some individual characteristic observed in both periods.[9,10] In addition, covariates must be observed in the post-treatment sample. Since covariates are often pre-treatment variables, this second requirement may prove a problem when covariates are time-varying and there are not retrospective covariate data available. Examples of applications of DID estimators with covariates to repeated cross sections are Card (1990, 1992), Meyer *et al*. (1995), Eissa and Liebman (1996), Acemoglu and Angrist (2001), Corak (2001) and Finkelstein (2002). Here, I show how to apply the methods proposed in this article to repeated cross sections. The data requirements are the same as for traditional difference-in-differences estimators which use cross-sectional data and covariates.

Assume that random samples are available for the pre-treatment and the post-treatment periods. For each individual in the pooled sample (post-treatment and pre-treatment), we observe $Z = (Y, D, T, X)$ where T is a time indicator that takes value one if the observation belongs to the post-treatment sample.

Assumption 3.3. *Conditional on* $T = 0$*, the data are* i.i.d. *from the distribution of* $(Y(0), D, X)$*; conditional on* $T = 1$*, the data are* i.i.d. *from the distribution of* $(Y(1), D, X)$.

This sampling scheme produces the following mixture distribution:

$$\begin{aligned} P_M(Y = y, D = d, X = x, T = t) &= \lambda \cdot t \cdot P(Y(1) = y, D = d, X = x) \\ &\quad + (1-\lambda) \cdot (1-t) \cdot P(Y(0) = y, D = d, X = x), \end{aligned}$$

where $\lambda \in (0, 1)$ reflects the proportion of the observations sampled in the post-treatment period.[11] Let $E_M[\cdot]$ denote expectations with respect to $P_M(\cdot)$.

Lemma 3.2. *If Assumptions* 3.1 *and* 3.3 *hold, and for values of* X *such that* $0 < P(D = 1 \mid X) < 1$*, we have* $E[Y^1(1) - Y^0(1) \mid X, D = 1] = E_M[\varphi_0 \cdot Y \mid X]$*, where*

$$\varphi_0 = \frac{T - \lambda}{\lambda \cdot (1-\lambda)} \cdot \frac{D - P(D = 1 \mid X)}{P(D = 1 \mid X) \cdot P(D = 0 \mid X)}.$$

9. Note that identification requires in turn that such individual characteristic is excluded from X. Otherwise, the support condition in Assumption 3.2 would be violated. This exclusion restriction is problematic if the excluded variable influences the dynamics of the outcome variable, so Assumption 3.1 is not plausible.

10. The requirement may also be satisfied in other cases, for example when the pre-treatment sample can be linked to administrative data records on treatment participation.

11. For simplicity, I do not consider more complicated situations in which the data may be generated by stratified sampling (on X or D). In such a case, the results in this section apply for a suitably reweighted sample (see, *e.g.* Wooldridge, 2002).

Then, following a reasoning similar to that of the previous section we have that the average treatment effect on the treated is identified by

$$E_M\left[\frac{P(D=1\mid X)}{P(D=1)}\cdot\varphi_0\cdot Y\right]=E[Y^1(1)-Y^0(1)\mid D=1]. \tag{12}$$

The following proposition is analogous to Proposition 3.1.

Proposition 3.2. *If Assumptions* 3.1*,* 3.2*, and* 3.3 *hold, then for* θ_0 *defined in equation* (11) *we have*

$$\theta_0=\arg\min_{\theta\in\Theta}E_M[P(D=1\mid X)\cdot\{\varphi_0\cdot Y-g(X_k;\theta)\}^2]. \tag{13}$$

The result in equation (12) can be obtained by considering a constant $g(X_k,\theta)$ in equation (13).

3.2.2. Selection on observables. In the previous section, it is shown how to approximate conditional average treatment effects by first weighting temporal differences in the outcome variable on the propensity score, and then projecting the weighted differences on a set of parametric functions of the covariates. The interpretation of the resulting functionals as approximations to conditional average treatment effects comes from the difference-in-differences condition in Assumption 3.1. However, it should be noticed that the same "first weight, then project" strategy can be applied in other contexts. In particular, the results in the previous section carry over naturally to "selection on observables":

$$E[Y^0(1)\mid X,D=1]=E[Y^0(1)\mid X,D=0]. \tag{14}$$

The reason is that "selection on observables" can be expressed as a particular case of Assumption 3.1 (when $E[Y^0(0)\mid X,D=1]=E[Y^0(0)\mid X,D=0]$). As a result, if equation (14) holds, then

$$\theta_0=\arg\min_{\theta\in\Theta}E[P(D=1\mid X)\cdot\{\rho_0\cdot Y-g(X_k;\theta)\}^2],$$

for θ_0 defined in equation (11) and $Y=Y(1)$.[12,13] Similar estimators have been considered in Wooldridge (2001). Imbens *et al*. (2003) consider the situation in which the average treatment effect is estimated for a given distribution of the covariates. Abadie (2003) applies similar approximation methods in the context of instrumental variable models for treatment effects.

3.2.3. Multilevel treatments. So far, we have considered only the case of a binary treatment, which is the usual focus of DID estimators. However, the same ideas can be applied when individuals may be exposed to different levels (or doses) of the treatment. Let W represent the level of the treatment. For untreated individuals, let $W=0$. For the treated, suppose that W takes on a finite number of positive values $w_1<\cdots<w_J$, with positive probability.[14] Let

12. For this case, it is useful to define $Y=Y(1)$ because equation (14) may be adopted as an identification restriction in absence of measures on the outcome variable in a pretreatment period. However, as shown for Ashenfelter's dip example, it may be necessary to condition on the values of the outcome variable in a pre-treatment period in order for equation (14) to hold.

13. Alternatively, if the objects of interest are average treatment effects conditional only on the covariates, θ_0 can be redefined as $\theta_0=\arg\min_{\theta\in\Theta}E[\{E[Y^1(1)-Y^0(1)\mid X_k]-g(X_k,\theta)\}^2]$. However, equation (14) does not identify the effect of the treatment on the untreated because it leaves $E[Y^1(1)\mid X,D=0]$ completely unrestricted. If we assume in addition that $E[Y^1(1)\mid X,D=1]=E[Y^1(1)\mid X,D=0]$, then θ_0 is identified by $\theta_0=\arg\min_{\theta\in\Theta}E[(\rho_0\cdot Y-g(X_k;\theta))^2]$.

14. Here, I assume that treatment levels are ordered (*e.g.* number of weeks in a training programme). For expositional simplicity and since it is often the case in applications, I consider only a finite number of treatment levels. However, the analysis presented in this section can be generalized to continuous treatments by substituting densities for W for probabilities for W, and integrals over those densities for sums.

$\mathcal{W}^+ = \{w_1, \ldots, w_J\}$, then $D = 1_{\mathcal{W}^+}(W)$, where $1_A(\cdot)$ is the indicator function for the set A (that is, $1_A(w) = 1$ if $w \in A$, zero otherwise). For $w \in \{0\} \cup \mathcal{W}^+$ and $t \in \{0, 1\}$, let $Y^w(t)$ be the potential outcome for treatment level w and period t. Suppose that Assumption 3.1 holds for each treatment level: $E[Y^0(1) - Y^0(0) \mid X, W = w] = E[Y^0(1) - Y^0(0) \mid X, W = 0]$, for $w \in \mathcal{W}^+$. In words, this assumption requires that, in absence of the treatment, the average outcomes for all treatment groups would have followed parallel trends, conditional on the covariates. As in the usual DID case with a binary treatment variable, the assumption allows the levels of average potential outcomes without the treatment to differ arbitrarily between treatment groups.

For $w \in \mathcal{W}^+$, let $\rho_0^w = (1_{\{w\}}(W)/P(W = w \mid X)) - (1_{\{0\}}(W)/P(W = 0 \mid X))$. (Note that, for ρ_0 defined in Lemma 3.1, $\rho_0 = \rho_0^1$.) Then, following the same reasoning as for Lemma 3.1, we obtain $E[\rho_0^w(Y(1) - Y(0)) \mid X] = E[Y^w(1) - Y^0(1) \mid X, W = w]$. In addition, for some class of square-integrable approximating functions $\mathcal{G} = \{g(W, X_k; \theta) : \theta \in \Theta\}$, redefine

$$\theta_0 = \arg\min_{\theta \in \Theta} E[(E[Y^W(1) - Y^0(1) \mid X_k, W] - g(W, X_k; \theta))^2 \mid D = 1]. \tag{15}$$

The parameters θ_0 define a least squares approximation to a function describing average effects for the treated $E[Y^w(1) - Y^0(1) \mid X_k, W = w]$. Then, these parameters are identified by

$$\theta_0 = \arg\min_{\theta \in \Theta} E\left[\sum_{j=1}^{J} P(W = w_j \mid X)(\rho_0^{w_j}(Y(1) - Y(0)) - g(w_j, X_k; \theta))^2\right].$$

This result collapses to the result in Proposition 3.1 if W is binary, and can be proven using the same argument as for Proposition 3.1.

4. ESTIMATION AND ASYMPTOTIC DISTRIBUTION

For concreteness, I will concentrate here on linear approximations to $E[Y^1(1) - Y^0(1) \mid X_k, D = 1]$ where X_k is a deterministic function of X. In addition, only the case of repeated cross sections is explicitly considered here. However, the analysis is also valid for the case of repeated observations if $Y(1) - Y(0)$ is substituted for $((T - \lambda)/\lambda(1 - \lambda)) \cdot Y$, and expectations are taken with respect to the distribution of $(Y(1), Y(0), D, X)$. Consider

$$\beta_0 = \arg\min_{\beta \in \Theta} E_M[\pi_0 \cdot \{\varphi_0 Y - X_k'\beta\}^2]$$

where $\pi_0(X) = P(D = 1 \mid X)$, and

$$\varphi_0 = \frac{T - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D - \pi_0(X)}{\pi_0(X) \cdot (1 - \pi_0(X))}.$$

Consider also the following estimator of β_0:

$$\widehat{\beta} = \left(\frac{1}{n}\sum_{i=1}^{n} X_{ki}\widehat{\pi}(X_i)X_{ki}'\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} X_{ki}\widehat{\pi}(X_i)\widehat{\varphi}_i Y_i,$$

where $\widehat{\pi}(X_i)$ is an estimator of $\pi_0(X_i)$, and

$$\widehat{\varphi}_i = \frac{T_i - \lambda}{\lambda \cdot (1 - \lambda)} \cdot \frac{D_i - \widehat{\pi}(X_i)}{\widehat{\pi}(X_i) \cdot (1 - \widehat{\pi}(X_i))},$$

for $\lambda = n_1/(n_0 + n_1)$. Under the conditions of the theorems stated below, $\widehat{\beta}$ is well defined with probability approaching one.

4.1. Non-parametric first step estimation of the propensity score

Here, I consider the case in which non-parametric (power series) regression is used in a first step to estimate π_0. Let $\zeta = (\zeta_1, \ldots, \zeta_r)'$ be a vector of non-negative integers where r is the dimension of X. Also let $X^\zeta = \prod_{j=1}^r X_j^{\zeta_j}$ and $|\zeta| = \sum_{j=1}^r \zeta_j$. Let $\{\zeta(k)\}_{k=1}^\infty$ be a sequence containing all distinct vectors ζ, with $|\zeta|$ non-decreasing. For a positive integer K, let $p^K(X) = (p_{1K}(X), \dots, p_{KK}(X))'$ where $p_{kK}(X) = X^{\zeta(k)}$. Then, for $K = K(n) \to \infty$ a power series non-parametric estimator of π_0 is given by

$$\widehat{\pi}(X) = p^K(X)'\widehat{\gamma} \tag{16}$$

where $\widehat{\gamma} = (\sum_{i=1}^n p^K(X_i)p^K(X_i)')^-(\sum_{i=1}^n p^K(X_i)D_i)$ and A^- denotes any symmetric generalized inverse of the matrix A.

Assumption 4.1. (i) $K^6/n = o(1)$, *$\pi_0(X)$ is continuously differentiable of order s, and* $nK^{-2s/r} = O(1)$; (ii) *the support of X is a Cartesian product of compact intervals on which X has density that is bounded away from zero;* (iii) *$\pi_0(X)$ is bounded away from zero and one;* (iv) *β_0 is an interior point of a compact set $\Theta \subset \mathbb{R}^k$;* (v) *$E_M Y^2 < \infty$, $\|X_k\|$ is bounded, and $E[X_k X_k' \mid D = 1]$ is non-singular.*

Let

$$\delta(X) = E_M \left[X_k \left(\frac{T-\lambda}{\lambda \cdot (1-\lambda)} \cdot \frac{D-1}{(1-\pi_0)^2} \cdot Y - X_k'\beta_0 \right) \middle| X \right].$$

Theorem 4.1. *If $n_0, n_1 \to \infty$, $n_1/(n_0+n_1) = \lambda \in (0,1)$ and Assumptions* 3.1, 3.3 *and* 4.1 *hold, then $n^{1/2}(\widehat{\beta} - \beta_0) \xrightarrow{d} N(0, V)$, where $V = Q^{-1}\Sigma Q^{-1}$, $Q = E[X_k D X_k']$, $\Sigma = E_M[\psi\psi']$, $\psi = X_k \pi_0(X)(\varphi_0 Y - X_k'\beta_0) + \delta(X) \cdot (D - \pi_0(X))$.*

To construct an estimator of the asymptotic variance, let $\widehat{V} = \widehat{Q}^{-1}\widehat{\Sigma}\widehat{Q}^{-1}$, where

$$\widehat{Q} = \frac{1}{n}\sum_{i=1}^n X_{ki} D_i X_{ki}', \qquad \widehat{\Sigma} = \frac{1}{n}\sum_{i=1}^n \widehat{\psi}_i \widehat{\psi}_i',$$

$$\widehat{\delta}(X_i) = \left(\sum_{i=1}^n X_{ki} \left(\frac{T_i - \lambda}{\lambda \cdot (1-\lambda)} \cdot \frac{D_i - 1}{(1-\widehat{\pi}(X_i))^2} \cdot Y_i - X_{ki}'\widehat{\beta} \right) p^K(X_i)' \right) \times \left(\sum_{i=1}^n p^K(X_i) p^K(X_i)' \right)^- p^K(X_i)$$

and $\widehat{\psi}_i = X_{ki}\widehat{\pi}(X_i)(\widehat{\varphi}_i Y_i - X_{ki}'\widehat{\beta}) + \widehat{\delta}(X_i) \cdot (D_i - \widehat{\pi}(X_i))$. The following theorem establishes the consistency of $\widehat{V}$.

Theorem 4.2. *If the assumptions of Theorem* 4.1 *hold and $K^7/n \to 0$, then $\widehat{V} \xrightarrow{p} V$.*

4.2. Parametric first step estimation of the propensity score

Often, samples are too small to use a non-parametric estimator in the first step. This is particularly likely when the analyst deals with longitudinal data-sets. In such cases, it may be convenient to use a parametric restriction in the first step and estimate the propensity score by maximum likelihood. This section provides distribution theory for that case. The results include probit, logit and linear probability first step estimation of π_0 as special cases.

Assumption 4.2. (i) *γ_0 is an interior point of a compact set $\Gamma \subset \mathbb{R}^r$;* (ii) *the support of X is a subset of a compact set S; $E[XX']$ is non-singular;* (iii) *there is a* (*known*) *function $\pi : \mathbb{R} \mapsto [0,1]$ such that $\pi_0(X) = \pi(X'\gamma_0)$;* (iv) *let $\mathcal{V} = \{x'\gamma : x \in S, \gamma \in \Gamma\}$; for $v \in \mathcal{V}$, $\pi(v)$ is bounded away from zero and one, strictly increasing and continuously differentiable with*